

LESLIE KADENGE

Data Scientist

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SUMMARY

Information Technology graduate working through a public portfolio of data science projects — supervised classification, regression, unsupervised learning and information retrieval, each reproducible from the linked repository by running one script.

The through-line is measuring honestly: establishing a baseline before claiming a result, testing whether an apparent improvement survives a significance test, and reporting the cases where it does not.

PROJECTS

Credit Card Fraud Detection Python · scikit-learn · FastAPI · Docker

github.com/les-k/fraud-detection

- 492 frauds in 284,807 transactions — 0.17%. Established up front that a model flagging nothing scores 99.87% accuracy, making accuracy unusable as a metric.
- Demonstrated that ROC-AUC and average precision select different models on this data, and that following average precision costs 12% more; selected on expected cost instead.
- Set the decision threshold from an explicit cost model, giving 0.05 rather than the 0.50 default, worth \$2,296 on its own.
- Result: 82% recall at 53% precision against a 0.17% base rate, recovering \$12,168 across a 48-hour holdout. Time-ordered train/test split, since a random split trains on the future.

Semantic Search Python · sentence-transformers · information retrieval

github.com/les-k/semantic-search

- Compared six retrieval methods over 1,400 documents against 225 queries and 1,837 human relevance judgments.
- Hybrid retrieval reached recall@10 of 0.437 against 0.377 for BM25 alone, and 0.010 for random ranking.
- Applied paired significance testing to every claimed improvement: two of five did not survive, including one with a higher average on every metric ($p = 0.895$).

Customer Segmentation Python · scikit-learn · SQL (DuckDB)

github.com/les-k/customer-segmentation

- K-means over 8,068 customers on mixed categorical and numeric features, with the same analysis written in both pandas and SQL and cross-checked to four decimal places.
- Reported that silhouette scores stayed below 0.25 at every k and that the clusters did not reproduce the existing segmentation (adjusted Rand index 0.079) — a negative result, stated as one.
- Holding spending out of the clustering entirely, demographic clusters still separated high spenders from 6.2% to 21.8% ($\chi^2 = 976$, $p < 0.001$).
- SQL covering joins, aggregation, window functions, CTEs and QUALIFY.

Wine Quality Prediction Python · scikit-learn · imbalanced classification

github.com/les-k/wine-quality

- Identified 11% duplicate rows leaking across the train/test split and inflating accuracy by 11 percentage points, with nothing in the metrics indicating it.
- Showed 58.4% accuracy against a macro-F1 of 0.279, with three of six quality grades scoring exactly zero on precision and recall.
- Reframed the task as a question the data could answer, reaching 0.756 macro-F1 against a baseline scoring 86.7% accuracy while finding no good wines at all.

Used Car Price Prediction Python · regression · cross-validation

github.com/les-k/car-price-prediction

- Established that a rule of thumb anyone could apply in their head — showroom price $\times$ 65.4% — already reaches R^2 0.555, then made every model addition justify itself against that rather than against zero.
- The full linear model reached R^2 0.769 and cut the average miss from 1.81 to 1.41 lakhs: the honest size of the gain, not 0.77 from nothing.

- bullet A random forest scored 0.540 on the test split but 0.856 ± 0.163 across 5-fold cross-validation — three times the spread of the linear model. The single split had caught a bad fold, not a worse model.
- bullet Three cars out of 62 accounted for 72% of the forest's squared error; on the other 59 it beat linear regression on mean absolute error, 1.29 against 1.49. The better model for the cars the data is mostly made of, and the worse one by R^2 — both true at once.
- bullet Converted year to age rather than using it directly, since a model trained on raw years learns that 2018 is expensive and breaks on the first 2020 car it sees.

Advertising Spend Analysis Python · statistical inference

github.com/les-k/sales-prediction

- bullet Computed coefficient standard errors and t-tests directly to test each channel's contribution.
- bullet Found newspaper spend indistinguishable from noise ($p = 0.72$) — it correlated with sales only through radio, which feature ranking by correlation would have missed.
- bullet Detected non-additivity in the residuals and added a TV $\times$ radio interaction term, taking test R-squared from 0.894 to 0.973 and halving RMSE.

TECHNICAL SKILLS

LANGUAGES Python, SQL

DATA pandas, NumPy, SciPy, DuckDB

MACHINE LEARNING scikit-learn, sentence-transformers; classification, regression, clustering, retrieval

STATISTICS Hypothesis testing, significance testing, cost-sensitive thresholds, experiment design

VISUALISATION matplotlib, seaborn, Tableau

ENGINEERING FastAPI, Docker, Git, GitHub Actions, pytest

EDUCATION

BSc Information Technology — Zetech University, November 2023 Second-class honours, upper division

Coursework: Machine Learning, Data Science, Artificial Intelligence, Database Systems, Software Engineering, Distributed Systems, Programming Fundamentals (Python, Java, C, C++, JavaScript).

CERTIFICATION

ICDL Profile — Computer Pride.

References available on request.